In Class Notes

V. Markos

2026-10-27

Last Time Recap

Last we briefly discussed, among others, key concepts around Random Variables, including:

- Probability Mass Function, i.e., the function $f_X(x) = P(X = x)$.
- Cumulative Distribution Function, i.e., the functions $F_X(x) = P(X \leq x)$.

We also discussed some key properties of the above:

- $0 \leq f_X(x) \leq 1$
- $f_X(x) = 0$ for x not in the support of X.
- $\sum_x f_X(x) = 1$ since we add on all possible outcomes.

Similarly, for the cdf:

- F_x is increasing with respect to x, since we add non-negative probabilities.
- $\lim_{x \to -\infty} F_X(x) = 0$
- $\lim_{x \to +\infty} F_X(x) = 1$
- $\lim_{x \to x_0^+} F_X(x) = F_X(x_0)$

Discrete Random Variables

Usually, a discrete random variable counts the count / number of something. We shall first provide an overview of the most well-known discrete random variables.

Well Known Discrete Random Variables

Bernoulli Random Variables

Consider a coin and let X be the number of H in a *single* toss of this coin. Evidently, $X \in \{0, 1\}$, i.e., the support of X is $\{0, 1\}$. So, in this case, $X = 0$ means T (no H), while $X = 1$ means H. Also, let $p \in [0, 1]$ is the probability that the coin yields H. So, we have:

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$

This is the so called *Bernoulli* random variable. Typically, a Bernoulli random variable is not that interesting itself but, as we shall see, it is the basis for many more complex and interesting random variables. In this case, the PMF reads:

$$f_X(x) = P(X = x) = \begin{cases} p & x = 1 \\ 1 - p & x = 0 \end{cases}.$$

Binomial Random Variable

Assume we have a coin that yields H with probability $p \in [0, 1]$. Let us now toss this coin n times independently, for some $n = 1, 2, \ldots$ Let also X, in this case, be the number of tosses that yield H. This is the so-called *Binomial* random variable, and its PMF is:

$$f_X(x) = P(X = x) = P(x \text{ tosses yield H}).$$

To compute the probability that x out of n tosses yield H, we think as follows:

- In order to get x times H, we have a probability of p^x.
- The rest tosses should yield T, which happens with probability $(1 - p)^{n-x}$.
- So, each combination of x H and $n - x$ T occurs with probability: $p^x(1 - p)^{n-x}$.
- It now remains to compute the ways in which x out n tosses can be H. This corresponds to choosing x out of n tosses to be H, i.e.:
$$\binom{n}{x}.$$

So, in total, we have:

$$f_X(x) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}.$$

Using probabilities, we often get algebraic identities for free. For example:

> Observe that, since we know that $\sum f_X(x) = 1$, i.e., the total probability is preserved. This means that for any $p \in [0, 1]$, we must have:
>
> $$\sum_{x=0}^{n} \binom{n}{x} p^x (1 - p)^{n-x} = 1.$$

Let us now consider an example. Consider a lithography factory that creates some microchips in which there is a particular machine that creates a chip of type A. This machine produces a faulty chip with probability $p = 1\%$. What is then, the probability that in a sample of $10,000$ chips, we get at most one faulty chip?

Let X be the number of faulty chips in a sample of $n = 10,000$ chips. Then, X follows a Binomial distribution with parameters $n = 10,000$ and $p = 0.01$, i.e., the probability that a chip is faulty. So, we are interested into:

$$P(X \leq 1) = P(X = 0) + P(X = 1).$$

Given then PMF of X, we have:

$$f_X(0) = P(X = 0) = \binom{10,000}{0} 0.01^0 0.99^{10,000} = 0.99^{10,000}.$$
$$f_X(1) = P(X = 1) = \binom{10,000}{1} 0.01^1 0.99^{9,999} = 10,000 \cdot 0.01 \cdot 0.99^{9,999} = 100 \cdot 0,99^{9,999}.$$

So, the requested probability is:

$$P(X \leq 1) = 0.99^{10,000} + 100 \cdot 0,99^{9,999}$$

Using R:

```
p <- 0.99^{10000} + 100*0.99^{9999}
print(p)
```

```
## [1] 2.293977e-42
```

Geometric Random Variable

Consider the following experiment:

> Let p be the probability that a coind yields H. What is the probability that after tossing the coin many times independently the first time that it yields H is after n tosses?

Let X be the number of consecutive Ts until the first time we get H. Then, the support of X is $\{0, 1, 2, \ldots\}$. So, we are interested into the probability:

$$f_X(x) = P(X = x) = P(x \text{ times T before the first H}).$$

For example, we are interested in cases like: TTTH, TH, H, TTTTTTTTH and, in general:

$$\underbrace{T \ldots T}_{x} H.$$

Since all tosses are independent, the probability for the above to happen is:

$$f_X(x) = P(X = x) = (1 - p)^x p, \quad x = 0, 1, \ldots$$

Observe again, that this also yields a known analytic series for free:

$$\sum_{x=0}^{\infty} f_X(x) = 1 \Leftrightarrow \sum_{x=0}^{\infty} (1 - p)^x p = 1,$$

that is, dividing both sides by p:

$$\sum_{x=0}^{\infty} (1 - p)^x = \frac{1}{p}.$$

As an example, let us consider a lottery where we pick among 20,000 different candidates. Further assume that each candidate participates always with the same number. So, what is the probability that the candidate with number 12,567 will win the lottery after at least 10 failures.

Let $p = 1/20,000$ be the probability that one wins the lottery, and also let X be the number of failures until one's first win. We are interested in the value:

$$P(X \geq 10).$$

There are two ways to compute this. One is to compute the infinite sum:

$$P(X \geq 10) = \sum_{x=10}^{\infty} P(X = x).$$

The other way is to use the number one trick we saw with the b-day paradox:

$$P(X \geq 10) = 1 - P(X < 10) = 1 - \sum_{x=0}^{9} P(X = x).$$

While both are relatively easy, the latter requires less advanced maths. Substituting the formula for $f_X(x)$ we have:

$$P(X \leq 10) = 1 - \sum_{x=0}^{9} \left(1 - \frac{1}{20000}\right)^x \frac{1}{20000} = 1 - \frac{1}{20000} \sum_{x=0}^{9} \left(\frac{19999}{20000}\right)^x$$

Here we can either use the known geometric sum formula:

$$\sum_{k=0}^{n} \lambda^k = \frac{1 - \lambda^{n+1}}{1 - \lambda}.$$

But, we also have R:

```
q <- 19999/20000
p <- 1 - q
r <- 1 - p * sum(q^(0:9))
print(r)
```

```
## [1] 0.9995001
```

This is expected as the chances to win the lottery are really low, so, if one is to win, this should be expected to take at least 10 iterations.

Negative Binomial Random Variable

Imagine that we toss a coin (independently) that yields H with probability p until we reach $r > 0$ Hs. We are now interested in the number of T until this happens. So, let X is this number. What is then, the probability:

$$f_X(x) = P(X = x), \quad x = 0, 1, 2, \ldots$$

With a rationale similar to that of the Binomial Distribution:

- We get r Hs with probability p^r.
- We get x Ts with probability $(1 - p)^x$.

- We need in total $r + x$ tosses, of which r should be H (and x should be T).

So, in a similar fashion:

$$P(X = x) = \binom{r+x}{r} p^r (1-p)^x.$$

But, if we visit, e.g., the Negative Binomial Wikipedia page we see that the correct formula should be:

$$P(X = x) = \binom{r+x-1}{x} p^r (1-p)^x.$$

There are two errors, both in the binomial coefficient. To begin with, the last toss will always be H, so we do not have $r + x$ but $r + x - 1$ choices (for the first $r + x - 1$ tosses). So, we should have written:

$$\binom{r+x-1}{r-1}$$

since we can choose the position of the first $r - 1$ Hs. Then, since, in general:

$$\binom{n}{k} = \binom{n}{n-k},$$

this also reads:

$$\binom{r+x-1}{r-1} = \binom{r+x-1}{x}.$$

This is intuitively sound, since, e.g., to pick $k = 7$ out of $n = 10$ balls (without replacement), it is the same as to pick the $n - k = 3$ balls that will *not* be picked.

So, to conclude:

$$f_X(x) = P(X = x) = \binom{r+x-1}{x} p^r (1-p)^x.$$

Poisson Random Variable

Consider the case of a typographer that makes, on average, 2 typos per (typed) page. In this case, the number X, of typos per page follows the so-called *Poisson* distributions, with PMF reading:

$$f_X(x) = P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!},$$

where λ is a parameter of the distribution, corresponding to the average value. In our case:

$$f_X(x) = e^{-2} \frac{2^x}{x!}, \quad x = 0, 1, \ldots$$

In the above, e is Euler's constant, i.e. $e \approx 2.71828\ldots$

Also, assuming that the above formula holds, we can derive a know Taylor expansion:

$$\sum_{x=0}^{\infty} f_X(x) = 1 \Leftrightarrow \sum_{x=0}^{\infty} e^{-\lambda} \frac{\lambda^x}{x!} = 1.$$

Hence, multiplying both side by e^{λ}, we get;

$$\sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{\lambda}.$$

Okay, But How to Distinguish Between Distributions?

Below we present some example cases for each distribution:

- Binary things, i.e., success / failure, are typically a Bernoulli.
- The count of successes in a finite series of trials is typically a Binomial.
- The count of failures up to the first success is typically a Geometric.
- The count of failures up to the first r successes is typically a Negative Binomial.
- Also, Poisson typically corresponds to these cases where we have some sort of "additivity" or the so-called "memory-less" property.